



CHI-SQUARE TEST FOR VARIANCE

HYPOTHESIS TESTING

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TOPIC OUTLINE

Chi-Square Test for Variance



CHI-SQUARE TEST FOR VARIANCE



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The chi-square test for variance is a statistical method that compares the sample variance to the hypothesized population variance.

Hypothesis

$$H_o: \sigma_1 = \sigma_o$$

$$H_a: \sigma_1 \neq \sigma_o \text{ (p-value} < \alpha \text{)}$$

Assumptions

- Continuous data
- Normal data and non-normal data

Test Statistic

$$\chi^2 = (n - 1) \frac{s^2}{\sigma^2}$$

where

n = sample size

s^2 = sample variance

σ^2 = population variance



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syntax

```
from scipy import stats
```

Left-Tailed Test

```
p_value = stats.chi2.cdf(  
    chi_sq_stat, dof)
```

Right-Tailed Test

```
p_value = 1-stats.chi2.cdf(  
    chi_sq_stat, dof)
```



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syntax

```
from scipy import stats
```

Two-Tailed Test

```
p_value = 2 * min(  
  
    stats.chi2.cdf(chi_sq_stat, dof),  
  
    1 - stats.chi2.cdf(chi_sq_stat, dof))
```



EXERCISE

The dataset contains the electricity production in MWh by the following production types:

Type	μ	σ
Nuclear	1283.78	32.50
Wind	779.86	88.28
Hydroelectric	1796.86	96.24
Oil and Gas	1160.69	60.66
Coal	1139.33	42.59
Solar	167.58	36.08
Biomass	55.12	2.13

dataset

electricity-normal-sample-cleaned.csv

Perform a chi-square test for variance to determine whether the variance of electricity production (in MWh) for each production type in the given dataset differs significantly from known population parameters.



LABORATORY

